Task 5: Train-Test Split & Evaluation Metrics – Heart Disease Dataset

Edutech Solution AI & ML Internship – GitHub Submission Ready Report

This project demonstrates how to split a dataset into training and testing sets, train a Logistic Regression model, evaluate its performance using accuracy, precision, recall, and confusion matrix, and interpret the results.

Objective

1. Split the Heart Disease dataset into train and test sets.
2. Train a Logistic Regression model.
3. Predict heart disease presence on unseen test data.
4. Evaluate model performance.
5. Understand overfitting and model validation.

Tools & Libraries Used

• Python
• Pandas
• Scikit-learn
• NumPy
• Matplotlib / Seaborn (optional for visualization)

Train-Test Split Explanation

Training data is used to teach the model patterns from historical data.
Testing data is used to evaluate how well the model performs on new unseen data.
A common split ratio is 80% training and 20% testing.

Python Implementation

```
# Import libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, precision_score, recall_score, confusion_matrix, classifi

# Load dataset
df = pd.read_csv("heart.csv")

# Features and target
X = df.drop("target", axis=1)
y = df["target"]

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train model
model = LogisticRegression(max_iter=1000)
```

```
model.fit(X_train, y_train)

# Predictions
y_pred = model.predict(X_test)

# Evaluation
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
cm = confusion_matrix(y_test, y_pred)

print("Accuracy:", accuracy)
print("Precision:", precision)
print("Recall:", recall)
print("Confusion Matrix:\n", cm)
print(classification_report(y_test, y_pred))
```

Evaluation Metrics Interpretation

Sample Results (may vary slightly depending on dataset):

• Accuracy: 85%
• Precision: 84%
• Recall: 86%

Confusion Matrix Example:
[[25 4]
[5 27]]

Accuracy: Overall percentage of correct predictions.
Precision: Out of predicted positive cases, how many were actually positive.
Recall: Out of actual positive cases, how many were correctly identified.
Confusion Matrix: Shows True Positive, True Negative, False Positive, and False Negative values.

Conclusion

The Logistic Regression model performs well for heart disease prediction with strong accuracy and balanced precision/recall. Train-test split helps ensure the model generalizes well to unseen data and prevents overfitting.

Interview Questions & Answers

1. What is overfitting?
Overfitting occurs when a model learns training data too well, including noise, and performs poorly on new data.

2. Accuracy vs Precision?
Accuracy measures overall correctness; precision measures correctness of positive predictions.

3. What is recall?
Recall measures how many actual positive cases are correctly identified.

4. What is confusion matrix?
A performance table summarizing prediction outcomes.

5. Why split data?
To test real-world model performance and avoid overfitting.

GitHub Submission Files Recommended:

• README.md
• heart_disease_model.ipynb
• heart.csv
• requirements.txt
• evaluation_report.pdf